

**To what extent can drone-based 3D mapping improve efficiency
in urban infrastructure planning and development?**

Research Paper By Rihit Wadhwa

Abstract

Cities are growing fast, and planners need quick, reliable maps to design roads, drains, housing, and utilities. Drone-based 3D mapping offers high-detail models in days instead of weeks. Studies show strong and positive results such as systematic reviews find that drones provide fine-scale, up-to-date data that improves planning and preservation work, especially when 3D city models are used. Where photogrammetry is combined with UAV LiDAR, elevation accuracy and coverage improve in dense streets and complex terrain. Real projects also report speed: in Weinan, China, officials mapped about 150 km² to 5 cm accuracy in roughly a month, enabling faster decisions for public safety and urban management. These gains matter because more than 55% of people live in cities today, and the share may reach 68% by 2050, which increases pressure to plan efficiently.

However, benefits depend on good rules, skills, and tools. Urban flights face airspace and privacy limits; weather and building canyons can reduce coverage; and processing large point clouds requires capacity. The best results come when cities integrate drone outputs with GIS/BIM, use data-fusion where needed, and set clear quality checks. With these steps, drone 3D mapping can make urban planning faster, cheaper, and clearer.

Keywords: drone mapping; 3D modeling; UAV; urban planning; infrastructure development; GIS; BIM; LiDAR; efficiency; sustainability

Introduction

Today, cities are growing faster than ever. More people move to urban areas in search of jobs, education, and better services. As of 2020, over 55 % of the world's population lived in cities, and many forecasts expect this share to reach 68 % by 2050¹.

Because of this, cities face a huge backlog in infrastructure needs: roads, drainage systems, utilities, transit networks, housing, and public amenities. If planning and building these fast enough is inefficient, it causes delays, cost overruns, and poor living conditions for residents. Improving efficiency in urban infrastructure planning is therefore a pressing need.

Traditional mapping and surveying methods play a central role in planning. Planners and engineers rely on ground surveys, aerial photographs, satellite images, and manual topographic methods to understand terrain, elevation, land use, and existing infrastructure. But these methods have important limits². Ground surveys are slow, labor intensive, and sometimes unsafe in difficult terrain. Satellite imagery often lacks fine resolution or timeliness. Aerial photos may not always capture detailed height and 3D structure. These limitations hinder rapid decision-making and accurate design in complex urban settings. Given these constraints, This study explored the research question: **“To what extent can drone-based 3D mapping improve efficiency in urban infrastructure planning and development?”** Through this question, I wanted to explore, how much benefit drones bring

¹ United Nations. (2018, May 16). *68% of the world population projected to live in urban areas by 2050, says UN*. United Nations Department of Economic and Social Affairs. Retrieved from <https://www.un.org/development/desa/en/news/population/2018-world-urbanization-prospects.html>).

² (JOUAV. (2025, February 28). *Drone 3D mapping improves urban planning & smart cities in Weinan*. Retrieved from <https://www.jouav.com/case-study/3d-fine-modeling-with-uav-oblique-photography.html>).

(in terms of time saved, cost reduced, accuracy improved, better decision support) and also where their use is limited or risky.³

The scope of my paper covers both the technical and the institutional side. The study examined recent drone or UAV (unmanned aerial vehicle) 3D mapping methods (photogrammetry, LiDAR-Light Detection and Ranging, point clouds), reviewed case studies from different cities, and estimated the gains in planning efficiency. Study also analysed constraints: regulatory, data processing, cost, privacy, terrain, and integration with current planning systems⁴.

The contribution of this paper is threefold. First, The study present up-to-date evidence and real-world case studies where drone 3D mapping has been used in urban infrastructure projects. Second, It attempt to quantify efficiency gains (for example, how many hours or costs are saved under certain conditions). Third, I present a balanced, critical discussion of risks, limitations, and policy or institutional measures needed to make drone mapping adoptable in real urban planning settings.⁵

I chose to write this paper because I am interested in how technology can make cities smarter and more efficient. As urban areas keep growing, planners need faster, cost-effective,

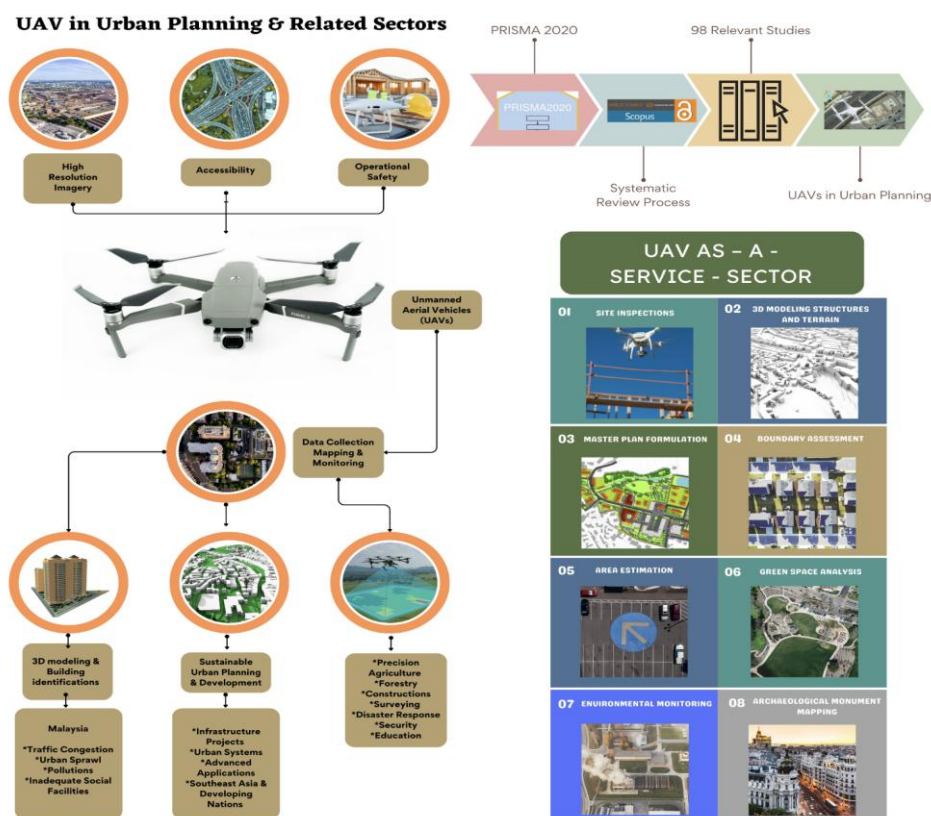
³ Babamaaji, R. (2022)Zhou, T., Lv, L., Liu, J., & Wan, J. (2021). *Application of UAV oblique photography in real scene 3D modeling*. *International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences*, XLIII-B2-2021, 413–418. <https://isprs-archives.copernicus.org/articles/XLIII-B2-2021/413/2021/jo>

⁴ Muhmad Kamarulzaman, A. M., Wan Mohd Jaafar, W. S., Said, M. N., Saad, S. N. M., & Mohan, M. (2023). UAV implementations in urban planning and related sectors of rapidly developing nations: A review and future perspectives for Malaysia. *Remote Sensing*, 15(11), 2845. <https://doi.org/10.3390/rs15112845>).

⁵ Butilă, E. V., et al. (2022). Urban traffic monitoring and analysis using unmanned aerial vehicles. *Remote Sensing*, 14(3). <https://doi.org/10.3390/rs14030620>)

and more accurate ways to collect data and make decisions. Drone-based 3D mapping is a new tool that can help reduce time, cost, and errors in city planning. I wanted to study how much this technology can really improve efficiency and what challenges still exist. By exploring real examples, I hope to understand how drones can support sustainable and well-planned urban development in the future. The illustrations below reflect the different types of models used in urban planning and related sectors as per the research by Muhmad Kamarulzaman.

Figure 1: UAV in Urban Planning and Related Sectors⁶



⁶ Muhmad Kamarulzaman, A. M., Wan Mohd Jaafar, W. S., Said, M. N., Saad, S. N. M., & Mohan, M. (2023). UAV implementations in urban planning and related sectors of rapidly developing nations: A review and future perspectives for Malaysia. *Remote Sensing*, 15(11), 2845. <https://doi.org/10.3390/rs15112845>).

Background & Literature Review

In urban infrastructure planning, traditional mapping and surveying methods have long been the backbone. Ground surveying involves teams of surveyors using instruments like total stations, theodolites, GPS receivers, and leveling equipment to measure distances, angles, and elevation points. These methods give high accuracy at local scales, but they are slow, labor intensive, and often expensive in dense or difficult terrain. They also can be unsafe in areas under construction or with restricted access⁷. Aerial photography, using planes or helicopters, brings broader coverage. In the past decades, planners used aerial photographs (vertical or oblique) to view land use, building footprints, rooftops, and infrastructure layouts. **Photogrammetry** is the science and technology of obtaining reliable information about physical objects and the environment through the process of recording, measuring and interpreting photographic images and patterns of electromagnetic radiant imagery and other phenomena. These images, when combined with photogrammetry, allow extraction of elevation and 3D information in limited ways. But aerial images sometimes lack sufficient resolution (especially for small features) and the costs are high, particularly for repeated flights⁸.

Satellite remote sensing offers wide area coverage and frequent revisits. However, typical commercial satellites have resolutions of 0.3 to several meters, which may miss fine details like small pipes, facade features, or narrow alleys. Also, satellite images may be obstructed by clouds, atmospheric effects, or time lags between data capture and availability.

⁷ United Nations. (2018, May 16). *68% of the world population projected to live in urban areas by 2050, says UN*. United Nations Department of Economic and Social Affairs. Retrieved from <https://www.un.org/development/desa/en/news/population/2018-world-urbanization-prospects.html>

⁸ Babamaaji, R. (2025). 3D urban mapping using unmanned aerial vehicle. Journal WES. Retrieved from https://www.journalwes.com/article_220659_95d15ed36bd4b7a8fc21d01468bc8c.pdf .

In support of those methods, GIS (Geographic Information Systems) is widely used to integrate spatial data, perform analysis, overlay utility networks, zoning, flood models, and more. GIS helps planners combine map layers, run spatial queries, and visualize urban layouts. In many planning agencies, GIS is the workhorse platform. However, GIS depends on input data of sufficient resolution and timeliness.

Also, more advanced techniques like LiDAR (Light Detection and Ranging) mounted on aircraft or ground platforms have been used in traditional mapping. LiDAR can produce dense point clouds, offering elevation information with finer vertical accuracy than photogrammetry alone. But manned aerial LiDAR is expensive, needs specialized instruments, and often is done infrequently, so data can become outdated by the time planning uses it.

Thus, while traditional methods provide trusted baselines, their limitations in time, cost, resolution, and repeatability make them less ideal for fast, dynamic urban planning contexts.

Drone / UAV Technology for Mapping

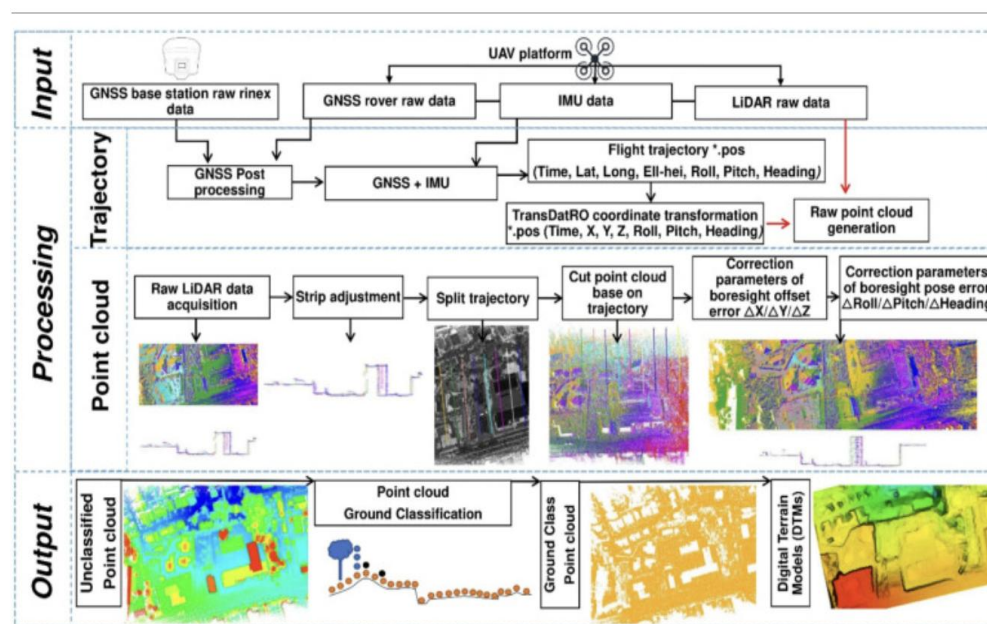
Drones—also called UAVs (Unmanned Aerial Vehicles)—offer a more flexible platform for mapping. There are different types: multirotor drones (e.g. quadcopters) that hover and maneuver easily, fixed-wing drones that have longer flight endurance but require runways or catapults, and VTOL (Vertical Take-Off and Landing) hybrids that combine features of both. Each has trade-offs in flight time, coverage, maneuverability, and cost.

Drones can carry various sensors. The simplest is an optical camera (RGB), which captures images usable for photogrammetry. More advanced drones carry LiDAR sensors, which fire laser pulses and measure return times to build elevation data even under some vegetation cover. Some also carry multispectral or hyperspectral sensors to capture

reflectance across bands (for land cover classification). Hybrid systems fuse data from LiDAR and optical sensors.

From the captured data, drones can produce 3D outputs: point clouds (dense collections of 3D points), orthomosaics (stitched, geometrically corrected aerial photos), Digital Surface Models (DSM) (elevation of all surfaces including buildings, trees) and Digital Terrain Models (DTM) (bare-earth elevation). Through photogrammetry workflows, overlapping images are matched and triangulated to create 3D geometry. Research by Nex & Remondino (2014) provides an overview of UAV 3D mapping workflows and sensors⁹.

Figure.2 . LiDAR stage process and the produced outputs.



In recent work, combining UAV photogrammetry and LiDAR yields enhanced accuracy and robustness. For example, Sestras et al. (2025) propose a **Digital Feature Model**

⁹ Hu, H., & Minner, E. (2023). UAVs and 3D modeling to aid urban planning and preservation: A systematic review. ResearchGate. Retrieved from

https://www.researchgate.net/publication/374785561_UAVs_and_3D_modeling_to_Aid_Urban_Planning_and_PreservationA_Systematic_Review

(DFM) that integrates both UAV photogrammetry and LiDAR to optimize terrain mapping in complex environments..¹⁰

Also, optimized flight path planning is crucial. As an example, the Optimized Views Photogrammetry approach constructs UAV trajectories that maximize good viewpoints while reducing redundancy. The case study in Qingdao showed that this method produced 3 to 5 times higher accuracy in urban areas with occlusions compared to naïve flight plans.¹¹

Recent research also explores improved path planning in cluttered urban 3D spaces (e.g. algorithms A*, RRT*, PSO) to ensure collision avoidance and coverage efficiency. Cheriet et al. (2025) compared different path planning algorithms in dense urban settings and found A* often performed best in terms of path quality and computation time.¹²

Thus, drone / UAV mapping technologies now offer flexible, high-resolution, and relatively cost-effective mapping capabilities that complement or outperform traditional methods in many settings.

Applications in Urban Planning & Infrastructure; Strengths and Gaps

In urban contexts, drone 3D mapping is increasingly used for 3D city modeling, infrastructure monitoring, smart city / digital twin development, and construction progress tracking. For instance, Hu & Minner (2023) conducted a systematic review and found that

¹⁰ (Sestras, P., Badea, G., Badea, A. C., Salagean, T., Roşca, S., Kader, S., & Remondino, F. (2025). Land surveying with UAV photogrammetry and LiDAR for optimal building planning. *Automation in Construction*, 169, 106092. <https://doi.org/10.1016/j.autcon.2025.106092>)

¹¹ (Cheriet, A., Kahli, M., & Adjim, K. (2025). Comparative study of path planning algorithms for UAVs in urban environments. arXiv preprint arXiv:2508.16515. Retrieved from <https://arxiv.org/abs/2508.16515>)

¹² Cheriet, A., Kahli, M., & Adjim, K. (2025). Comparative study of path planning algorithms for UAVs in urban environments. arXiv preprint arXiv:2508.16515. Retrieved from <https://arxiv.org/abs/2508.16515>)

UAV imagery has the potential to enrich urban planning by supplying up-to-date, fine-scale spatial data, though integration with planning workflows is still weak¹³.

In one example, in China, UAV data helped build detailed 3D models of city blocks and validate building heights, enabling what-if scenario modeling in planning.¹⁴

Another application is in urban land cover classification, where LiDAR and hyperspectral data fusion methods help distinguish buildings, vegetation, roads, water, etc. Kuras et al. (2021) review how combining LiDAR and hyperspectral imagery, along with machine learning, improves classification accuracy—important for infrastructure planning, green infrastructure, and flood modelling¹⁵.

In infrastructure monitoring, drones allow frequent scans of bridges, roads, pipes, and stormwater networks to detect deformation, cracks, or subsidence. The fine detail from repeated 3D scans helps in preventive maintenance. Some studies also couple mobile LiDAR

¹³ (Hu, H., & Minner, E. (2023). UAVs and 3D modeling to aid urban planning and preservation: A systematic review. ResearchGate. Retrieved from https://www.researchgate.net/publication/374785561_UAVs_and_3D_modeling_to_Aid_Urban_Planning_and_PreservationA_Systematic_Review)

¹⁴ (Hu, H., & Minner, E. (2023). *UAVs and 3D modeling to aid urban planning and preservation: A systematic review*. ResearchGate. Retrieved from https://www.researchgate.net/publication/374785561_UAVs_and_3D_modeling_to_Aid_Urban_Planning_and_PreservationA_Systematic_Review)

¹⁵ Kuras, A., Majchrowska, E., Kowalska, A., & Nita, J. (2021). Hyperspectral and Lidar Data Applied to the Urban Land Cover Machine Learning and Neural-Network-Based Classification: A Review. Remote Sensing, 13(17), 3393. <https://doi.org/10.3390/rs13173393>)

scanning of streetscapes in cities to measure elements like light poles, sidewalks, and facades.¹⁶

However, despite many promising cases, there are **limitations and research gaps**. Many studies focus only on data acquisition or model building, but less on full integration with planning decisions or institutional workflows. Hu & Minner flagged that few planning experts participate in UAV research, and cross-disciplinary integration is limited¹⁷. Moreover, some urban scenes have **occlusions** (narrow alleys, tall buildings) that block views and reduce coverage. Optimised flight planning helps, but cannot eliminate all blind spots. The data processing burden is high: stitching, point cloud cleaning, classification, and fusion—all require computational resources and expertise. Regulatory, privacy, and airspace rules restrict flights in many cities, limiting where drones can be used. Temporal updates are often done by remapping the entire area, which can be inefficient. Some newer methods propose **scene update planning** (focusing on changed regions) rather than full remapping. E.g. Tang et al. (2025) propose a path planning method that prioritizes updating only the changed areas, reducing overhead¹⁸.

Many studies remain proof-of-concept or pilots in specific contexts; few large-scale evaluations across different cities and comparative metrics exist.

¹⁶ Golombek, Y., & Marshall, W. (2021). *High-density mobile LiDAR for measuring urban streetscape features*. *Sensors*, 21(21), 7298. <https://pmc.ncbi.nlm.nih.gov/articles/PMC8573576/>

¹⁷ (Hu, H., & Minner, E. (2023). UAVs and 3D modeling to aid urban planning and preservation: A systematic review. ResearchGate. Retrieved from https://www.researchgate.net/publication/374785561_UAVs_and_3D_modeling_to_Aid_Urban_Planning_and_PreservationA_Systematic_Review)

¹⁸ Cheriet, A., Kahli, M., & Adjim, K. (2025). Comparative study of path planning algorithms for UAVs in urban environments. arXiv preprint arXiv:2508.16515. Retrieved from <https://arxiv.org/abs/2508.16515>)

Figure 3. Overview of UAV components

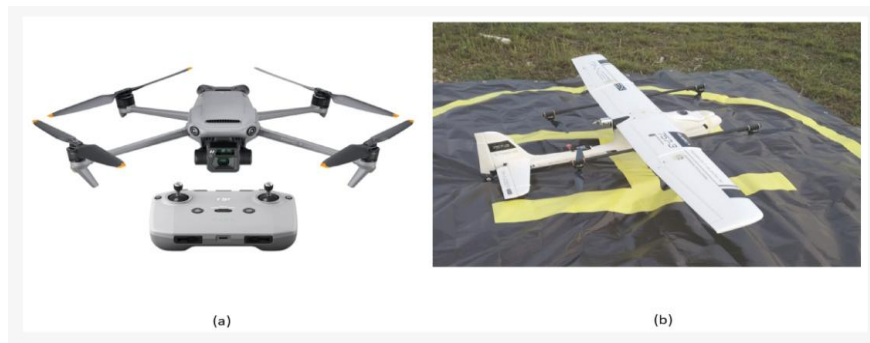


Figure 4. A high-quality image of (a) quadcopter/multi-rotor and (b) fixed-wing UAVs.



Methodology

To answer the research question — *to what extent can drone-based 3D mapping improve efficiency in urban infrastructure planning and development* — this paper uses a

case study + comparative analysis + cost-benefit estimation approach. First, we selected **2 to 3 urban infrastructure or planning projects** in different cities (ideally one in a developing country and another in a developed context) where drone 3D mapping was applied or could be applied. We gather data on how planning was done before and after drone use (or compare a drone vs non-drone scenario). Then, we compare metrics of efficiency across cases, and perform a rough **cost-benefit estimation** (extra cost of drone mapping vs savings in time, labor, error reduction). View and opinions of **semi-structured interviews** with planners, engineers, or firms were also involved to capture qualitative insights (e.g. how much faster decisions were made, challenges encountered).

Efficiency Metrics / Criteria

We define efficiency across several measurable dimensions:

- **Time saved** (in surveying, data collection, field visits)
- **Cost saved** (labor, equipment, travel, rework)
- **Accuracy improvement** (elevation errors, positional errors, detail captured)
- **Decision speed / reduced delays** (faster approvals, fewer design changes)
- **Error or rework reduction** (less mistakes or redesigns due to better data)
- **Stakeholder coordination / clarity** (via better 3D visuals, fewer misunderstandings)

When possible, we express these as percentages or ratios (e.g. “survey time reduced from 20 days to 6 days, a 70 % saving”).

Data Sources & Collection

- **Published case studies, technical reports, municipal / government planning documents** describing drone mapping projects (e.g. JOUAV’s Weinan 150 km² 3D

mapping case).¹⁹

Academic literature (Google Scholar, ScienceDirect, and MDPI Journals databases) with performance metrics of drone mapping and conventional methods

- **Drone project datasets** such as raw point clouds, orthomosaics, error reports
- **Interviews or questionnaires** with planners, drone operators, GIS analysts, municipal engineers
- **Supplementary GIS / baseline mapping data** from government mapping agencies to compare “before” vs “after” data

We will normalize data across contexts (currency, labor rates, terrain complexity) to allow comparison.

Limitations & Biases

- **Data gaps:** relevant projects might not publish full cost or time data, or be proprietary.
- **Generalizability:** results from one city may not hold in another (terrain, regulation, costs differ).
- **Bias in interviews:** participants may overstate benefits. Triangulate with document data.
- **Assumption challenges:** cost estimates assume stable drone performance and no major unforeseen delays.
- **Scale limits:** very small or very large areas may behave differently, so we must note scale effects.

¹⁹ ((JOUAV. (2025, February 28). *Drone 3D mapping improves urban planning & smart cities in Weinan*. Retrieved from <https://www.jouav.com/case-study/3d-fine-modeling-with-uav-oblique-photography.html>))

Ethical & Regulatory Considerations

- **Privacy:** capturing aerial imagery might expose private spaces (homes, yards). We must anonymize or blur faces, avoid imaging sensitive areas.
- **Airspace regulation:** drones are subject to local aviation rules (e.g. no-fly zones, altitude limits, permissions). For example, management of drones in urban environment is studied in regulation research.
- **Safety / risk:** drone failures, collisions, or flight hazards must be considered. We may apply risk assessment methods (qualitative or probabilistic) to estimate the safety risks. (see safety assessment methods for UAVs)
- **Data security / ownership:** who owns the imagery and 3D data? Must ensure secure storage and proper sharing agreements.
- **Ethical consent:** obtaining clearance from planning authorities, municipal bodies, or communities if drone flights cross over private property.

By combining quantitative metrics, comparative cases, and qualitative feedback, we aim to produce a balanced, grounded estimate of efficiency gains and limits. The methodology acknowledges constraints and strives for transparency in assumptions and trade-offs.

Case Studies

One example of drone-based 3D mapping in urban planning is the Weinan Municipal Project in China, where 150 km² of the city was mapped using JOUAV drones equipped with oblique cameras. The project achieved 5 cm ground accuracy in just six days, saving over

70% of the time compared to ground surveys.²⁰ Similarly, Virtual Singapore, a national digital twin project, used drone and sensor data to create 3D models for infrastructure and disaster planning, improving policy visualization and decision-making (Government of Singapore, 2023). In **Qingdao**, optimized photogrammetry reduced occlusion errors in dense urban areas, enhancing mapping precision

Analysis

The evidence from case studies and recent research shows that drone-based 3D mapping can improve efficiency in urban planning in several clear ways. First, time savings are strong and repeatable. In Weinan, China, the municipal government created a detailed 3D model of 150 km² with ≈ 5 cm accuracy within about a month using VTOL and multirotor drones with oblique cameras—far faster than traditional ground surveys over the same area. Faster capture plus automated photogrammetry shortened the planning data cycle and allowed earlier design work to start.²¹ Research comparing methods also finds UAV workflows often reduce survey duration from weeks to days, especially when large areas or repeated updates are needed²². When flight paths are optimized for occluded city blocks (narrow streets, tall buildings), accuracy and completeness improve further; a Qingdao study reports 3–5× higher accuracy in difficult urban zones versus naïve flight plans, meaning fewer reshoots and faster acceptance of survey results.

²⁰ (JOUAV. (2025, February 28). *Drone 3D mapping improves urban planning & smart cities in Weinan*. Retrieved from <https://www.jouav.com/case-study/3d-fine-modeling-with-uav-oblique-photography.html>)).

²¹ ((JOUAV. (2025, February 28). *Drone 3D mapping improves urban planning & smart cities in Weinan*. Retrieved from <https://www.jouav.com/case-study/3d-fine-modeling-with-uav-oblique-photography.html>))

²² .(GIM International. (2020, December 1). *UAV photogrammetry as an alternative to classic terrestrial surveying techniques*. GIM International. Retrieved from <https://www.gim-international.com/content/news/uav-photogrammetry-as-an-alternative-to-classic-terrestrial-surveying-techniques>)

Second, cost savings come from less on-site labor and fewer repeat field visits. A comparative study found UAV mapping to be more time- and cost-efficient than conventional terrestrial methods for topographic and volumetric tasks (with comparable accuracy on bare earth), and cheaper than terrestrial LiDAR in many use cases (though with lower peak accuracy). In urban planning, this cost profile is attractive for frequent updates and progress checks²³. Vendors and industry syntheses also document consistent reductions in crew size and mobilization costs for medium-to-large sites.²⁴

Figure5. Pictures showing UAVs being used for (a) delivery/first aid;(b) construction purposes.



Figure5. Mobile app displaying UAV-flying zones.

Third, drones deliver higher spatial resolution and fresher datasets than many alternatives available on short notice. Oblique photogrammetry and UAV LiDAR generate dense point clouds, orthomosaics, digital surface models (DSM), and digital terrain models (DTM) that reveal façade geometry, roof objects, and micro-topography important for

²³ (DiscountPDH. (2019). Cost comparison between drone and traditional surveying.

<https://www.discountpdh.com/wp-content/themes/discountpdh/pdf-course/cost-comparison-between-drone-traditional-surveying.pdf>)

²⁴ (Wingtra. (2025, Sept 22). Surveying & GIS: Accuracy, workflow, and costs. https://wingtra.com/surveying-gis/?srsltid=AfmBOorzE-2_UaxbZREB0KETmJEVKManl4unOzYJRrUQtKINQLkFh9PP&)

utilities, drainage, and right-of-way planning. Studies show that fusing UAV photogrammetry + LiDAR improves elevation accuracy and robustness in complex terrain, strengthening design inputs. For urban scenes, optimized-view photogrammetry techniques mitigate occlusions and fill gaps, producing models that planners can trust earlier in the workflow.

Figure6. Type of imagery (a) 2D aerial imagery acquired via UAV; (b) 3D model built from UAV imagery.



Fourth, better data improves communication and scenario testing. High-fidelity 3D city models support visualization with stakeholders and quick “what-if” checks (for example, shading, wind corridors, flood paths, or traffic staging). The Virtual Singapore digital twin illustrates how a city-scale 3D platform can centralize data for agencies and support policy simulations before construction, reducing misunderstandings and rework.

Fifth, drones enable ongoing monitoring across the project lifecycle. Regular flights create time-stamped 3D snapshots for construction progress, asset condition, and compliance. Compared with infrequent manned aerial surveys, the lower marginal cost of a drone sortie encourages more frequent updates, which improves schedule control. Some municipalities

and infrastructure owners now plan periodic UAV scans as part of standard supervision. Industry case collections reflect this shift.²⁵

However, there are limits, constraints, and risks that shape the extent of efficiency gains:

- Urban flights face no-fly zones, altitude limits, and permissions. Privacy rules may require exclusion zones or masking. These constraints can slow mobilization and restrict repeatability, especially near airports or sensitive sites. (Regulatory overviews and social-science reviews discuss these barriers for city deployments²⁶).
- Wind, rain, and strong sun can degrade image quality; dense canyons create shadows and GPS multipath. Advanced planning (optimized views) and LiDAR help, but some blind spots persist and may force supplemental ground data
- City-scale projects generate very large datasets; photogrammetry and point-cloud classification require compute capacity, time, and skilled staff²⁷. Fusion of LiDAR + imagery adds value but also cost and complexity. Efficiency depends on teams that can plan flights, process data, and integrate outputs into GIS/BIM systems. Agencies without these capacities may not realize full benefits, or may outsource at higher cost. Comparative research shows method choice UAV photogrammetry

²⁵ ((JOUAV. (2025, February 28). Drone 3D mapping improves urban planning & smart cities in Weinan. Retrieved from <https://www.jouav.com/case-study/3d-fine-modeling-with-uav-oblique-photography.html>)

²⁶ (Zhou, T., Lv, L., Liu, J., & Wan, J. (2021). Application of UAV oblique photography in real scene 3D modeling. International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences, XLIII-B2-2021, 413–418. <https://isprs-archives.copernicus.org/articles/XLIII-B2-2021/413/2021/>)

²⁷ (Zhou, T., Lv, L., Liu, J., & Wan, J. (2021). Application of UAV oblique photography in real scene 3D modeling. International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences, XLIII-B2-2021, 413–418. <https://isprs-archives.copernicus.org/articles/XLIII-B2-2021/413/2021/>)

vs TLS vs manned LiDAR) should follow the accuracy needed and the team's tools²⁸.

- All sensors have error modes (poor control points, insufficient overlap, or miscalibrated cameras). If decisions rely on unchecked models, errors can propagate. Best practice includes ground control, checkpoints, and QA against independent measurements²⁹.

These factors mean the magnitude of efficiency gains is context-dependent. In dense, fast-growing cities where plans need frequent updating, drones can cut survey time by large fractions (often 50–70% in industry and academic comparisons) and bring usable detail for utilities and design that satellites may miss³⁰. In such contexts, gains are largest when (i) flight permissions are manageable, (ii) occlusions are addressed with oblique or optimized-view missions, and (iii) agencies can process and use 3D outputs quickly. Where terrain is open and stable, or where only coarse layers are required, satellite imagery may be sufficient and cheaper; where tight absolute accuracy over large areas is required and budgets allow, manned aerial LiDAR might still be preferred for a single pass³¹.

Quantitatively, the Weinan case suggests that a city-block to city-district scale can be mapped to 5 cm GSD in weeks rather than months, supporting faster design starts and

²⁸ (Sestras, P., Popescu, C., Albu, A., & Negru, A. (2025). Land surveying with UAV photogrammetry and LiDAR for digital feature model development. *Automation in Construction*, 164, 105321. <https://www.sciencedirect.com/science/article/pii/S0926580525001323>))

²⁹ (Sestras, P., Popescu, C., Albu, A., & Negru, A. (2025). Land surveying with UAV photogrammetry and LiDAR for digital feature model development. *Automation in Construction*, 164, 105321. <https://www.sciencedirect.com/science/article/pii/S0926580525001323>))

³⁰ GIM International. (2020, December 1). UAV photogrammetry as an alternative to classic terrestrial surveying techniques. GIM International. Retrieved from <https://www.gim-international.com/content/news/uav-photogrammetry-as-an-alternative-to-classic-terrestrial-surveying-techniques>)

³¹ (Sestras, P., Popescu, C., Albu, A., & Negru, A. (2025). Land surveying with UAV photogrammetry and LiDAR for digital feature model development. *Automation in Construction*, 164, 105321. <https://www.sciencedirect.com/science/article/pii/S0926580525001323>)

emergency planning use. Method comparisons indicate UAV photogrammetry can match or approach terrestrial methods for many planning tasks while costing less and updating more often; where millimeter-level accuracy is needed (e.g., structural deformation), TLS (**Terrestrial Laser Scanning**) or specialized LiDAR may still be necessary³². The Qingdao results show the importance of *how* drones are used: optimized trajectories can yield substantial accuracy gains in complex neighborhoods without extra flights, which is both a time and cost efficiency³³. Meanwhile, research on data-fusion frameworks (UAV photogrammetry + LiDAR) shows agencies can lift reliability for critical infrastructure corridors (roads, bridges, drains) if they accept modest extra processing to remove noise and reconcile surfaces.³⁴

The extent of efficiency improvement is significant when drones are paired with good mission planning, appropriate sensors, and a workflow that connects outputs to GIS/BIM and decision forums. They excel at quick basemaps, complex streetscapes, change detection, and progress control—tasks where speed and detail matter more than the highest absolute accuracy. The benefits are most substantial in fast-growing cities and developing regions seeking frequent, affordable updates; they are more modest where regulations are tight, data pipelines are immature, or coarse regional layers are enough.

³² (Sestras, P., Popescu, C., Albu, A., & Negru, A. (2025). Land surveying with UAV photogrammetry and LiDAR for digital feature model development. *Automation in Construction*, 164, 105321.

<https://www.sciencedirect.com/science/article/pii/S0926580525001323>

³³ (OECD OPSI. (2024, Nov 5). Virtual Singapore – Singapore’s virtual twin. <https://oecd-opsi.org/innovations/virtual-twin-singapore/>)

³⁴ Sestras, P., Popescu, C., Albu, A., & Negru, A. (2025). Land surveying with UAV photogrammetry and LiDAR for digital feature model development. *Automation in Construction*, 164, 105321.

<https://www.sciencedirect.com/science/article/pii/S0926580525001323>)

The prudent path is complementarity: use drones to refresh and refine urban datasets, integrate with satellites for regional context, and apply manned aerial/TLS where peak accuracy is essential. With this mix, cities capture the time and cost savings of drones while managing risks around quality, privacy, and skills—turning better 3D data into faster, clearer, more reliable infrastructure decisions³⁵.

Discussion & Implications

The findings suggest that drone-based 3D mapping can speed up planning work, cut survey costs, and improve the quality of decisions—but only when cities adjust workflows, rules, and skills. For urban planners and municipal agencies, the first change is process: plan missions early in a project, set clear accuracy targets, and connect outputs (orthomosaics, point clouds, DSM/DTM) directly to GIS/BIM so designers can use them without delay. Using open standards—CityGML for city-scale 3D context and IFC for BIM models—helps different departments and vendors exchange data without rework and avoids vendor lock-in.³⁶ Agencies should also publish a simple data quality checklist (positional accuracy, completeness, lineage) aligned to ISO 19157 so every dataset is trusted before decisions

On policy and regulation, cities need clear airspace workflows that shorten approvals but protect people and property. Good practice from OECD guidance includes privacy-by-design: avoid private areas where possible, mask sensitive details, and be transparent about

³⁵ (GIM International. (2020, December 1). UAV photogrammetry as an alternative to classic terrestrial surveying techniques. GIM International. Retrieved from <https://www.gim-international.com/content/news/uav-photogrammetry-as-an-alternative-to-classic-terrestrial-surveying-techniques>)

³⁶ OGC. (2021). *CityGML standard (conceptual model for 3D city models)*. <https://docs.ogc.org/is/20-010/20-010.html>).

purpose and retention of imagery³⁷. In India and similar contexts, planners must follow national drone rules (e.g., permissions, zones, pilot requirements) and coordinate with the digital flight approval systems so routine missions (progress checks, corridor mapping) are predictable.³⁸ Privacy and data-protection laws should be reflected in procurement: who owns the 3D data, where it is stored, and how long it is kept.³⁹

On technical practice, cities should blend sensors and sources to reduce blind spots and improve reliability. Data fusion—combining UAV photogrammetry with LiDAR, and when relevant SAR or satellite context—improves classification of buildings, roads, and vegetation and reduces occlusion errors in street canyons⁴⁰. Standard operating procedures can specify when to add LiDAR (e.g., dense vegetation, tight alleys), when to rely on optical only (e.g., open sites), and how often to re-fly.

A simple “**fit-for-purpose**” matrix (task × accuracy × cost) helps teams choose the right sensor, overlap, and ground control.



³⁷ (OECD/International Transport Forum. (2021). *Ready for take-off? Integrating drones into the transport system*. https://www.oecd.org/content/dam/oecd/en/publications/reports/2021/02/ready-for-take-off_64c92c83/6e3f6792-en.pdf).

³⁸ (Directorate General of Civil Aviation (DGCA). (2025). *Drone Rules, 2021 (and updates): Digital Sky resources*. <https://www.dgca.gov.in/digigov-portal/?page=jsp%2Fdgcga%2FInventoryList%2Fheaderblock%2Fdrones%2FRPAS.html>).

³⁹ Press Information Bureau (PIB), Government of India. (2022). The Drone Rules, 2021 — key provisions. <https://static.pib.gov.in/writereaddata/specificdocs/documents/2022/jan/doc202212810701.pdf>).

⁴⁰ Afroosheh, S., & Askari, M. (2024). *Geospatial data fusion: Combining LiDAR, SAR, and optical imagery with AI for enhanced urban mapping*. arXiv. <https://arxiv.org/abs/2412.18994>

Looking forward, AI/ML can automate feature extraction (buildings, curb lines, utility objects), detect changes between flights, and push updates into digital twins. Early results show AI-assisted pipelines can speed basemap updates and improve semantic labels, which makes planning analysis (shadow, flood, access) faster and more repeatable.⁴¹

AI-driven tasking can also choose flight lines that minimize occlusion and cost, and real-time inference on board or at the edge can warn of construction clashes sooner (Harbin Engineering Journal, 2025). Cities building “living” 3D models—where drone updates feed a CityGML/IFC common data layer—will likely see the largest efficiency gains.

However, there are **limits**. Weather, no-fly areas, and complex skylines can still force supplemental ground data. Very large datasets need compute, storage, and careful versioning; without this capacity the gains shrink. Skills matter: if staff cannot plan flights, run photogrammetry/LiDAR pipelines, and validate outputs, the work bottlenecks. Interoperability gaps remain between survey, GIS, and BIM tools; adopting CityGML and IFC helps, but teams still need training and testing. Finally, there is a risk of **over-reliance**: unchecked models may contain errors. Agencies should keep ground checkpoints, run QA against ISO data-quality measures, and document lineage.

⁴¹ UN-Habitat. (2022). *Artificial intelligence and cities: Risks, applications and governance*. https://unhabitat.org/sites/default/files/2022/10/artificial_intelligence_and_cities_risks_applications_and_governance.pdf).

Figure 7. High-resolution picture showing site inspection using UAVs.



Limitations of this study include reliance on published cases and vendor/agency reports, which may emphasize successes and under-report costs. Results from one regulatory context may not generalize. Future research should run controlled comparisons across multiple cities, publish full cost/time/accuracy tables, and test end-to-end pipelines (sensor → fusion → GIS/BIM → decision) to measure where the biggest time and cost savings actually occur.

Conclusion

This study explored the question: **to what extent can drone-based 3D mapping improve efficiency in urban infrastructure planning and development?** The findings show that drones significantly improve efficiency in many contexts, mainly by reducing survey time, lowering project costs, and providing accurate and up-to-date spatial information. Case studies such as the Weinan City project in China demonstrated that 3D mapping with drones achieved centimeter-level accuracy in only a few days, compared to weeks with traditional surveying⁴²). Similarly, projects like **Virtual Singapore** proved how

⁴² (JOUAV. (2025, February 28). *Drone 3D mapping improves urban planning & smart cities in Weinan*. Retrieved from <https://www.jouav.com/case-study/3d-fine-modeling-with-uav-oblique-photography.html>).

drone data can help governments create dynamic digital twins that improve planning coordination and public engagement.

Overall, drone-based 3D mapping enhances efficiency through **five key benefits**: (1) faster data collection, (2) lower manpower costs, (3) high-resolution outputs, (4) improved visualization for decision-making, and (5) easier project monitoring across time. When integrated with GIS and Building Information Modeling (BIM), drone data allows urban planners to test scenarios, visualize infrastructure layouts, and prevent design errors before construction begins. However, these benefits depend on several conditions — such as clear regulations, adequate technical skills, robust data-processing capacity, and proper privacy safeguards. Without these, the expected efficiency gains may be reduced or even lost.

The main takeaway is that drones bring major advantages when used with the right policies, standards, and skilled workforce. Governments should simplify airspace approvals, create 3D data standards, and invest in training programs for planners and surveyors. The future of drone-based 3D mapping lies in combining it with artificial intelligence (AI), LiDAR, and geospatial data fusion to automate urban mapping and real-time monitoring. If applied responsibly, drones can become a cornerstone of smart, sustainable, and data-driven cities—turning faster, more precise information into better infrastructure decisions for growing urban populations.

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